Applied Math

Some things I felt were interesting enough to write down during grad school

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Contents

1 Spin coordinates **3**
1.1 Cartesian . . . 3
1.2 Cylindrical . . . 4

2 Matrix shennanegins **6**
2.1 Matrix group immersions . . . 6
2.1.1 The rotation group $SO(3)$ . . . 8

3 Old notes **11**
3.1 Stokeslet . . . 11
3.2 Quaternions . . . 11
3.3 Magnus expansion . . . 12

4 Hat map stuff **14**

5 Geometry **15**
5.1 complex curvatures . . . 15
5.2 Geodesic flow . . . 15

6 Numerical methods **17**
6.1 local error . . . 17
6.2 Extrapolation . . . 17
6.3 Root finding . . . 17
6.4 Lie group methods . . . 18

7 Arnold–Beltrami–Childress flow **19**

8 Algebraic geometry **21**

9 Differential forms **21**
9.1 Level sets, adapted frames . . . 22

10 Functional analysis and spectral methods **23**
10.1 Green's functions . . . 23
10.2 Leray projection . . . 23

11 Perturbation theory **24**
11.1 The various sizes of 'oh' . . . 24

12 'The' Calculus **25**
12.1 Differential . . . 25
12.2 The Del operator ∇ . . . 27
12.3 Integration . . . 29
12.4 Invariant integration . . . 30
12.5 Fourier stuff . . . 32
12.6 Lagrange multipliers . . . 34
12.7 Some boring theorems, definitions, technical stuff, etc... . . . 35
12.8 Rotations and other Lie group stuff . . . 35
12.9 Curvilinear Coordinates . . . 37

13 Constrained optimization **38**
13.1 Notes from Marsden . . . 38

14 Differential equations **40**
14.1 Stokes Equation . . . 40
14.2 Poisson equation . . . 40
14.3 Poisson equation . . . 40
14.4 Wave equation . . . 41
14.5 Screened Poisson equation . . . 42
14.6 Navier-Stokes equations . . . 42
14.6.1 Stokes equations (Low Reynolds number) . . . 43
14.7 Sturm-Liouville theory . . . 43
14.8 Associated Laguerre equation . . . 44
14.8.1 Laplace's Equation, Associated Legendre Polynomials, and Spherical Harmonics . . . 44
14.9 Symmetries of differential equations and Lie groups . . . 45

15 Special Functions **46**
15.1 Chebyshev Polynomials . . . 46
15.2 Hermite Polynomials . . . 46
15.3 Associated Legendre polynomials . . . 47
15.4 Spherical harmonics . . . 47
15.5 Bessel functions . . . 49

1 Spin coordinates

1.1 Cartesian

$$\boldsymbol{e}_x = \partial_x \boldsymbol{r} \tag{1}$$

$$\boldsymbol{e}_y = \partial_y \boldsymbol{r} \tag{2}$$

$$\boldsymbol{e}_z = \partial_z \boldsymbol{r} \tag{3}$$

$$\mathrm{d}\boldsymbol{r} = \boldsymbol{e}_x \,\mathrm{d}x + \boldsymbol{e}_y \,\mathrm{d}y + \boldsymbol{e}_z \,\mathrm{d}z \tag{4}$$

Let $s = x + iy, s^* = x - iy$ so

$$\partial_s \boldsymbol{r} = \boldsymbol{e}_x - i\boldsymbol{e}_y \tag{5}$$

$$\partial_{s^*} \boldsymbol{r} = \boldsymbol{e}_x + i\boldsymbol{e}_y \tag{6}$$

$$\mathrm{d}\boldsymbol{r} = \frac{1}{2}\partial_s \boldsymbol{r} \,\mathrm{d}s + \frac{1}{2}\partial_{s^*} \boldsymbol{r} \,\mathrm{d}s^* + \partial_z \boldsymbol{r} \,\mathrm{d}z \tag{7}$$

$$\boldsymbol{e}_+ = \frac{1}{\sqrt{2}}(\boldsymbol{e}_x - i\boldsymbol{e}_y) \tag{8}$$

$$\boldsymbol{e}_- = \frac{1}{\sqrt{2}}(\boldsymbol{e}_x + i\boldsymbol{e}_y) \tag{9}$$

$$\boldsymbol{e}_0 = \boldsymbol{e}_z \tag{10}$$

$$\boldsymbol{e}_x = \frac{1}{\sqrt{2}}(\boldsymbol{e}_+ + \boldsymbol{e}_-) \tag{11}$$

$$\boldsymbol{e}_y = \frac{i}{\sqrt{2}}(\boldsymbol{e}_+ - \boldsymbol{e}_-) \tag{12}$$

$$\boldsymbol{e}_z = \boldsymbol{e}_0 \tag{13}$$

$$\langle \boldsymbol{e}_m | \boldsymbol{e}_{m'} \rangle = \delta_{mm'} \tag{14}$$

$$\boldsymbol{e}_0 \times \boldsymbol{e}_\pm = \pm i \boldsymbol{e}_\pm \tag{15}$$

$$\boldsymbol{e}_+ \times \boldsymbol{e}_- = i \boldsymbol{e}_0 \tag{16}$$

$$\hat{\mathbf{e}}_0 \boldsymbol{e}_p = i p \boldsymbol{e}_p \tag{17}$$

1.2 Cylindrical

$$\boldsymbol{e}_\rho = \cos(\phi)\boldsymbol{e}_x + \sin(\phi)\boldsymbol{e}_y \tag{18}$$

$$= \frac{1}{\sqrt{2}}\left(e^{i\phi}\boldsymbol{e}_+ + e^{-i\phi}\boldsymbol{e}_-\right) \tag{19}$$

$$\boldsymbol{e}_\phi = -\sin(\phi)\boldsymbol{e}_x + \cos(\phi)\boldsymbol{e}_y \tag{20}$$

$$= \frac{i}{\sqrt{2}}\left(e^{i\phi}\boldsymbol{e}_+ - e^{-i\phi}\boldsymbol{e}_-\right) \tag{21}$$

- Coordinate functions

$$\rho = \sqrt{x^2 + y^2} \tag{22}$$

$$\theta = \tag{23}$$

$$\phi = \tag{24}$$

- Frame

$$\boldsymbol{e}_\rho = \cos(\phi)\boldsymbol{e}_x + \sin(\phi)\boldsymbol{e}_y \tag{25}$$

- Position vector

$$\boldsymbol{r} = \rho\boldsymbol{e}_\rho \tag{26}$$

2 Matrix shennanegins

2.1 Matrix group immersions

Let $e = (e_1, \ldots, e_N)$ be an orthonormal basis for real vector space $X \sim \mathbb{R}^N$ and let $F = (F_1, \ldots F_{N^2})$ be a basis for $M(X) \sim \mathbb{R}^{N^2}$, the space of linear operators on X. We will impose conditions on F later to make it adapted to a submanifold of $M(X)$ which we will define below, but for now we aren't making any assumptions about it other than that its a basis. Define the basis $E = (E_1, \ldots, E_{N^2})$ by

$$E_\alpha = e_i e_j^T \tag{27}$$

$$\alpha(i, j) = (j-1)N + i \tag{28}$$

If we write α_N for the remainder of α mod N then $0 \leq \alpha_N \leq N-1$ and there exists a unique $0 \leq n \leq N-1$

$$0 \leq \alpha_N \leq N-1 \tag{29}$$

$$\alpha = nN + \alpha_N \tag{30}$$

$$\alpha(i, j) = (j-1)N + i \tag{31}$$

$$i(\alpha) = \alpha_N \tag{32}$$

$$j(\alpha) = \frac{\alpha - \alpha_N}{N} + 1 \tag{33}$$

$$e_i e_j^T = a_{ij}^\beta E_\beta \tag{34}$$

$$E_\beta = a_\beta^{ij} e_i e_j^T \tag{35}$$

$$a_{ij}^{\beta} = \begin{cases} 1, & \beta = (j-1)N + i \\ 0, & \end{cases} \tag{36}$$

$$a_{\alpha}^{k\ell} = \begin{cases} 1, & \alpha = (\ell-1)N + i \\ 0, & \end{cases} \tag{37}$$

$$\tag{38}$$

$$\mathrm{tr}\left(\left(e_i e_j^T\right)^T e_k e_\ell^T\right) = \sum_{\alpha} a_{ij}^{\alpha} a_{k\ell}^{\alpha} = \delta_{ik}\delta_{j\ell} \tag{39}$$

$$\mathrm{tr}\left(E_\alpha^T E_\beta\right) = \sum_{ij} a_\alpha^{ij} a_\beta^{ij} = \delta_{\alpha\beta} \tag{40}$$

Say you have an operator that acts on E_α's like

$$G \cdot E_\beta = G_{\alpha\beta} E_\alpha \tag{41}$$

$$G \cdot e_i e_j^T = G_{k\ell,ij} e_k e_\ell^T \tag{42}$$

where $G_{\alpha\beta} = G_{\beta\alpha}$.

$$G \cdot e_i e_j^T = a_{ij}^{\beta} G \cdot E_\beta = a_{ij}^{\beta} G_{\alpha\beta} E_\alpha = a_{ij}^{\beta} a_\alpha^{k\ell} G_{\alpha\beta} e_k e_\ell^T \tag{43}$$

$$G_{k\ell,ij} = a^{\alpha}_{k\ell} G_{\alpha\beta} a^{\beta}_{ij} \tag{44}$$

$$G_{\alpha\beta} = a^{k\ell}_{\alpha} G_{k\ell,ij} a^{ij}_{\alpha} \tag{45}$$

2.1.1 The rotation group $SO(3)$

Suppose you find yourself in possesion of a parameterization of $SO(3)$,

$$Q : M \to SO(3) \subset M(3,\mathbb{R}) \sim \mathbb{R}^9, \tag{46}$$

by some manifold M. For example, by the group of unit quaternions

$$Sp(1) = \left\{ x \in \mathbb{H} | \, x^{\dagger} x = 1 \right\} \tag{47}$$

$$q = q_0 + \boldsymbol{q} = q_0 + q_k \boldsymbol{e}_k = q_0 + q_1 i + q_2 j + q_3 k \tag{48}$$

Spin(1)

$$Q(q) = \left(q_0^2 - |\boldsymbol{q}|^2\right) I + 2 \left|\boldsymbol{q}\right\rangle \left\langle \boldsymbol{q}\right| + 2 q_0 \hat{\boldsymbol{q}} \tag{49}$$

$$= \left(\left(q_0^2 - |\boldsymbol{q}|^2\right) \delta_{ij} + 2 q_i q_j + 2 q_0 \varepsilon_{ikj} q_k\right) |\boldsymbol{e}_i\rangle \left\langle \boldsymbol{e}_j\right| \tag{50}$$

$$\mathrm{tr}(Q) = 3 q_0^2 - |\boldsymbol{q}|^2 \tag{51}$$

$$= 4 q_0^2 - 1 \tag{52}$$

$$= 3 - 4 |\boldsymbol{q}|^2 \tag{53}$$

Proof:

$$
\begin{aligned}
Q_{\ell k} &= \langle \boldsymbol{e}_\ell | Q | \boldsymbol{e}_k \rangle && (54)\\
&= \left\langle \boldsymbol{e}_\ell \middle| q \boldsymbol{e}_k q^\dagger \right\rangle && (55)\\
&= \left\langle \boldsymbol{e}_\ell \middle| \left(2 q_k q_i + (q_0^2 - |\boldsymbol{q}|^2)\delta_{ki} + 2\varepsilon_{ijk} q_0 q_j\right) \boldsymbol{e}_i \right\rangle && (56)\\
&= 2 q_k q_\ell + (q_0^2 - |\boldsymbol{q}|^2)\delta_{\ell k} + \varepsilon_{\ell jk} q_0 q_j && (57)
\end{aligned}
$$

□

If you don't want to worry about numerical issues stuff screwing with the normalization, you can just divide by $q^\dagger q = q_0^2 + |\boldsymbol{q}|^2$

$$
\begin{aligned}
Q(q_0, q_1, q_2, q_3) &= \frac{\left(q_0^2 - |\boldsymbol{q}|^2\right) I + 2 \left|\boldsymbol{q}\right\rangle \left\langle \boldsymbol{q}\right| + 2 q_0 \hat{\boldsymbol{q}}}{q_0^2 + |\boldsymbol{q}|^2} && (58)\\
&= \frac{\left(q_0^2 - |\boldsymbol{q}|^2\right) \delta_{ij} + 2 q_i q_j + 2 q_0 \varepsilon_{ikj} q_k}{q_0^2 + |\boldsymbol{q}|^2} \left|\boldsymbol{e}_i\right\rangle \left\langle \boldsymbol{e}_j\right| && (59)
\end{aligned}
$$

$$
\operatorname{tr}(Q) = \frac{3 q_0^2 - |\boldsymbol{q}|^2}{q_0^2 + |\boldsymbol{q}|^2} \tag{60}
$$

$$
Q(q_0, \boldsymbol{q})^T = Q(q_0, \boldsymbol{q})^{-1} = Q(q_0, -\boldsymbol{q}) \tag{61}
$$

Axis-Angle

$$
\begin{aligned}
Q(\alpha, \theta, \phi) &= \exp(\alpha \hat{\boldsymbol{u}}) && (62)\\
&= |\boldsymbol{u}\rangle \langle \boldsymbol{u}| + \cos(\alpha) \left(I - |\boldsymbol{u}\rangle \langle \boldsymbol{u}|\right) + \sin(\alpha) \hat{\boldsymbol{u}} && (63)\\
&= |\boldsymbol{u}\rangle \langle \boldsymbol{u}| + \left(\cos^2\left(\frac{\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)\right) \left(I - |\boldsymbol{u}\rangle \langle \boldsymbol{u}|\right) + 2\cos\left(\frac{\alpha}{2}\right)\sin\left(\frac{\alpha}{2}\right) \hat{\boldsymbol{u}} && (64)
\end{aligned}
$$

where $\boldsymbol{u} = \boldsymbol{u}(\theta,\phi) = \sin(\theta)\cos(\phi)\boldsymbol{e}_1 + \sin(\theta)\sin(\phi)\boldsymbol{e}_2 + \cos(\theta)\boldsymbol{e}_3$ is a unit vector.

$$\mathrm{tr}(Q) = 1 + 2\cos(\alpha) \tag{65}$$

Proof: $q = \cos\left(\frac{\alpha}{2}\right) + \sin\left(\frac{\alpha}{2}\right)\boldsymbol{u}$, some trig identities, etc... □

Euler angles

$$Q(\alpha_1,\alpha_2,\alpha_3) = \exp(\alpha_3\hat{\boldsymbol{u}}_{\boldsymbol{3}})\exp(\alpha_2\hat{\boldsymbol{u}}_{\boldsymbol{2}})\exp(\alpha_1\hat{\boldsymbol{u}}_{\boldsymbol{1}}) \tag{66}$$

Direction cosines

$$Q = \sum_{i,j}\cos\left(\alpha_{ij}\right)|\boldsymbol{e}_i\rangle\langle\boldsymbol{e}_j| \tag{67}$$

where $\cos\left(\alpha_{ij}\right) = \left\langle \boldsymbol{e}_i \middle| \boldsymbol{e}'_j \right\rangle$ where $\boldsymbol{e}'_j$ is the rotated frame.

$$\mathrm{tr}(Q) = \cos(\alpha_{11}) + \cos(\alpha_{22}) + \cos(\alpha_{22}) \tag{68}$$

SU(2)

$$Q = \tag{69}$$

Cayley transform

Note: Cayley transform, exponential map, euler angles, axis-angle are all just maps of the form

$$Q : \mathfrak{so}(3) \to SO(3) \tag{70}$$

3 Old notes

3.1 Stokeslet

with force $\boldsymbol{F}$ creates flow

$$\boldsymbol{u} = \boldsymbol{F} \cdot J \tag{71}$$

$$p = \frac{\boldsymbol{F} \cdot \boldsymbol{r}}{4\pi r^3} \tag{72}$$

where

$$J = \frac{1}{8\pi\mu r}\left(\mathbb{1} + \frac{\boldsymbol{r}\boldsymbol{r}}{r^2}\right) \tag{73}$$

3.2 Quaternions

Rotations can be represented by unit quaternions:

$$q = q_0 + \boldsymbol{q} = q_0 + q_1 i + q_2 j + q_3 k \tag{74}$$

$$|q|^2 = 1 \tag{75}$$

$$= q q^\dagger = q^\dagger q = (q_0)^2 + \boldsymbol{q} \cdot \boldsymbol{q} \tag{76}$$

$$R R^T = R^T R = 1 \tag{77}$$

$$R' = \hat{\boldsymbol{\omega}} R = R \hat{\boldsymbol{\omega}}_0 \tag{78}$$

$$q' = \frac{1}{2}\boldsymbol{\omega} q = \frac{1}{2} q \boldsymbol{\omega}_0 \tag{79}$$

where

$$\boldsymbol{\omega}_0 = 2q^\dagger q' = \omega_1 i + \omega_2 j \omega_3 k \tag{80}$$

$$\boldsymbol{\omega} = 2q' q^\dagger = q(\omega_1 i + \omega_2 j + \omega_3 k) q^\dagger \tag{81}$$

are pure quaternions

Proof: $qq^\dagger = 1 \Rightarrow q'q^\dagger + qq'^\dagger = q'q^\dagger + (qq')^\dagger = 0$ □

$$\begin{bmatrix} 0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = 2 \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \\ -q_1 & q_0 & q_3 & -q_2 \\ -q_2 & -q_3 & q_0 & q_1 \\ -q_3 & q_2 & -q_1 & q_0 \end{bmatrix} \begin{bmatrix} q_0' \\ q_1' \\ q_2' \\ q_3' \end{bmatrix} \tag{82}$$

I feel like maybe there is a sign error in that matrix somewhere...

3.3 Magnus expansion

Let Y be a curve through a group of matrices G with Lie algebra $\mathfrak{g}$

$$Y' = A(t) Y_0, \qquad t \geq 0, \qquad Y(0) = Y_0 \tag{83}$$

solution:

$$Y(t) = \exp(\Theta(t)) Y_0 \tag{84}$$

Where the Θ satisfies

$$\Theta' = \mathrm{d}\exp_{\Theta}^{-1} A = \sum_{k=0}^{\infty} \frac{B_k}{k!} \mathrm{ad}_{\Theta}^{k} A \tag{85}$$

$$\Theta(0) = 0 \tag{86}$$

B_k are Betti numbers and $B_k = 0$ whenever $k > 1$ is odd.

Derivative of exponential map

$$\exp : \mathfrak{g} \to G \tag{87}$$

4 Hat map stuff

$$\nabla \times \boldsymbol{A} = \hat{\nabla} \cdot \boldsymbol{A} = \nabla \cdot \hat{\boldsymbol{A}} \tag{88}$$

5 Geometry

5.1 complex curvatures

add small imaginary perturbation to curatvatures to deal with cases where curvatures are equal or zero.

$$\kappa_1, \kappa_2 \mapsto \kappa_1 + i\epsilon, \kappa_2 - i\epsilon \tag{89}$$

$$K_{ij} \mapsto K_{ij} + i\epsilon J_{ij} \tag{90}$$

where J_{ij} is an antisymmetric, real operator tangent to surface

$$\overline{\kappa}, \kappa_G \mapsto \overline{\kappa}, \kappa_G + \epsilon^2 \tag{91}$$

5.2 Geodesic flow

A geodesic between p and q is a critical point of the length functional

$$L[\gamma] = \int_I \sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}\,\mathrm{d}t \tag{92}$$

subject to $\gamma(\partial I) = \{p, q\}$ where $I \subset \mathbb{R}$ is some connected interval. The First variation of the length funcional is

$$\delta L = \left.\frac{g_{ij}\dot{\gamma}^i\delta\gamma^j}{\sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}}\right|_{\partial I} + \int_I \left[\frac{\dot{\gamma}^i\dot{\gamma}^k\partial_j g_{ik}}{2\sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{g_{ij}\dot{\gamma}^i}{\sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}}\right)\right]\delta\gamma^j\,\mathrm{d}t \tag{93}$$

and the EL equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{g_{ij}\dot{\gamma}^i}{\sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}}\right) - \frac{\dot{\gamma}^i\dot{\gamma}^k\partial_j g_{ik}}{2\sqrt{g_{ij}\dot{\gamma}^i\dot{\gamma}^j}} = 0. \tag{94}$$

After reparametrizing in terms of arclength using

$$\frac{\mathrm{d}}{\mathrm{d}t} = \|\dot{\gamma}\| \frac{\mathrm{d}}{\mathrm{d}s}, \tag{95}$$

$$\gamma'^i = \frac{1}{\|\dot{\gamma}\|} \dot{\gamma}^i, \tag{96}$$

and contracting with the inverse metric g^{ij}, we can write the geodesic equation

$$\boxed{\gamma''^i + \frac{1}{2} g^{ij} \partial_k g_{jl} \gamma'^k \gamma'^l = 0} \tag{97}$$

Or in terms of Christoffel symbols

$$\boxed{\gamma''^i + g^{ij} g_{ml} \Gamma^m_{kj} \gamma'^k \gamma'^l = 0} \tag{98}$$

Let $x^i(0)$ and $\dot{x}^i(0)$ be the initial position and initial velocity of a geodesic expressed in a some local coordinate system. Then in a neighborhood of its' starting position, the geodesic satisfies:

$$x^i(t) = x^i(0) + t\dot{x}^i(0) - \frac{t^2}{2} \Gamma^i_{jk} \dot{x}^j(0) \dot{x}^k(0) + O(t^3) \tag{99}$$

6 Numerical methods

6.1 local error

Local truncation error is the contribution to error in each step of your algorithm. A pth-order accurate method with step size h means that the error is proportional to h^p.

6.2 Extrapolation

If you know $\theta^{[j]} = \theta(t_j) + O(h^p)$ for $j = \ldots, k-2, k-1$. Then

$$\sum_{i=1}^{p} a_i \theta^{[k-i]} = \theta(t_k) + O(h^p) \tag{100}$$

where the a's are found by solving

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ (-1)^1 & (-2)^1 & \cdots & (-p)^1 \\ (-1)^2 & (-2)^2 & \cdots & (-p)^2 \\ \vdots & \vdots & \ddots & \vdots \\ (-1)^{p-1} & (-2)^{p-1} & \cdots & (-p)^{p-1} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{101}$$

or $\sum_{i=1}^{p} (-i)^\alpha a_i = \delta_{\alpha 0}$ for $\alpha = 0, \ldots, p-1$.

6.3 Root finding

$$\boldsymbol{F}(\boldsymbol{x}^*) = 0 \tag{102}$$

$$\boldsymbol{F}(\boldsymbol{x}_\alpha) + \delta \boldsymbol{F}(\boldsymbol{x}_\alpha) \cdot (\boldsymbol{x}_{\alpha+1} - \boldsymbol{x}_\alpha) = O(|\boldsymbol{x}_{\alpha+1} - \boldsymbol{x}_\alpha|^2) \tag{103}$$

Define the sequence

$$\boldsymbol{x}_{\alpha+1} = \boldsymbol{x}_\alpha - [\delta \boldsymbol{F}(\boldsymbol{x}_\alpha)]^{-1} \cdot \boldsymbol{F}(\boldsymbol{x}_\alpha) \tag{104}$$

or in local coordinates

$$x_{k,\alpha+1} = x_{k,\alpha} - \delta F^{-1}_{km,\alpha} F_{m,\alpha} \tag{105}$$

Newton's method can be adapted to spaces with curvature in the obvious way by replacing partial derivatives with covariant ones.

6.4 Lie group methods

Matrix groups can be thought of as suubmanifolds of Euclidean space by identifying $GL(n,\mathbb{R}) \sim \mathbb{R}^{n^2}$. Let $x(t), y(t) = x(t) + O(h^p) \in G$ be the actual and computed solutions of some ode at time t. Standard numerical methods introduce errors which, in addition to making the computed value different from the actual value, result in a matrix which is not actually an element of the G. An example of this occurs when you apply forward Euler to the problem of computing the motion of a rigid body subject to external stresses. To describe the motion you need to know the trajectories of 3 non-colinear material points on the body $a, b, c \in B$. Then, $|a-b|, |b-c|, |c-a|$ should all be constant. The group multiplication of $SE(3)$ gives you a way to navigate the configuration space of the rigid body without changing $|a-b|, |b-c|, |c-a|$.

7 Arnold–Beltrami–Childress flow

$$\boldsymbol{v} = \begin{bmatrix} \sin(z) & 0 & \cos(y) \\ \cos(z) & \sin(x) & 0 \\ 0 & \cos(x) & \sin(y) \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} \tag{106}$$

$$= \begin{bmatrix} 0 & \sin(z) & \cos(y) \\ \cos(z) & 0 & \sin(x) \\ \sin(y) & \cos(x) & 0 \end{bmatrix} \begin{bmatrix} A \\ C \\ B \end{bmatrix} \tag{107}$$

$$\boldsymbol{w} = \begin{bmatrix} 0 & \cos(z) & \sin(y) \\ \sin(z) & 0 & \cos(x) \\ \cos(y) & \sin(x) & 0 \end{bmatrix} \begin{bmatrix} A \\ C \\ B \end{bmatrix} \tag{108}$$

$$\boldsymbol{v} + i\,\boldsymbol{w} = \begin{bmatrix} 0 & i\exp(-iz) & \exp(iy) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp(-iy) & \exp(ix) & 0 \end{bmatrix} \begin{bmatrix} A \\ C \\ B \end{bmatrix} \tag{109}$$

$$K = \begin{bmatrix} 0 & i\exp(-iz) & \exp(iy) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp(-iy) & \exp(ix) & 0 \end{bmatrix} \tag{110}$$

$$\nabla K = \boldsymbol{E}^x \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \exp(-ix) \\ 0 & i\exp(ix) & 0 \end{bmatrix} \tag{111}$$

$$+\boldsymbol{E}^y \begin{bmatrix} 0 & i\exp(-iz) & \exp\left(iy\right) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp\left(-iy\right) & \exp(ix) & 0 \end{bmatrix} \tag{112}$$

$$+\boldsymbol{E}^z \begin{bmatrix} 0 & i\exp(-iz) & \exp\left(iy\right) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp\left(-iy\right) & \exp(ix) & 0 \end{bmatrix} \tag{113}$$

$$= \boldsymbol{E}^x \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \exp(-ix) \\ 0 & i\exp(ix) & 0 \end{bmatrix} \tag{114}$$

$$+\boldsymbol{E}^y \begin{bmatrix} 0 & i\exp(-iz) & \exp\left(iy\right) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp\left(-iy\right) & \exp(ix) & 0 \end{bmatrix} \tag{115}$$

$$+\boldsymbol{E}^z \begin{bmatrix} 0 & i\exp(-iz) & \exp\left(iy\right) \\ \exp(iz) & 0 & i\exp(-ix) \\ i\exp\left(-iy\right) & \exp(ix) & 0 \end{bmatrix} \tag{116}$$

$$K^\dagger = \begin{bmatrix} 0 & \exp(-iz) & -i\exp\left(iy\right) \\ \exp(iz) & 0 & \exp(-ix) \\ \exp\left(-iy\right) & -i\exp(ix) & 0 \end{bmatrix} \tag{117}$$

$$= -iK \tag{118}$$

$$= \exp(-i\pi/2)K \tag{119}$$

$H = \exp(-i\pi/4)K$ Hermitian!

$$\left(\exp(-i\pi/4)K\right)^\dagger = \exp(i\pi/4)\exp(-i\pi/2)K \tag{120}$$

$$= \exp(-i\pi/4)K \tag{121}$$

$$\mathrm{tr}(H) = 0 \tag{122}$$

$$\det(H) = e^{-i3\pi/4}\left(-ie^{-iz}(e^{-ix-iy}) + e^{iy}(e^{ix+iz})\right) \tag{123}$$

$$= e^{-i3\pi/4}\left((e^{ix+iy+iz} - ie^{-ix-iy-iz}\right) \tag{124}$$

$$= e^{-i3\pi/4}\left((e^{i(x+y+z)} - ie^{-i(x+y+z)}\right) \tag{125}$$

$$\tag{126}$$

8 Algebraic geometry

An algebraic variety is the set of solutions to a system of polynomials. If you also allow for inequalities you can approximate submanifolds of Euclidean space with arbitrary precision. This probably has some fun applications in 3d printing or something. Let $q_{ijk}^{[M]} \in \mathbb{E}^3$ for $i, j, k \in \{0, ..., M\}$ be points and $p_\ell^{[M]} : \mathbb{E}^3 \to \mathbb{R}$ for x

9 Differential forms

Let ω be a k-form and $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_k, \boldsymbol{e}_{k+1}$ some tangent vectors at x and let $U(\epsilon)$ be the 'curvilinear parallelipiped' with vertices $x, x + \epsilon\boldsymbol{e}_1, \ldots, x + \epsilon\boldsymbol{e}_{k+1}$

$$F(\epsilon) = \int_{\partial U(\epsilon)} \omega = \mathrm{d}\omega[\boldsymbol{e}_1, \ldots, \boldsymbol{e}_k, \boldsymbol{e}_{k+1}] + O(\epsilon^2) \tag{127}$$

9.1 Level sets, adapted frames

$$\star \mathrm{d}\hat{\eta} = \frac{1}{2}\varepsilon^{i}_{\cdot jk} \nabla_i \hat{\eta}\, \mathrm{d}x^j \wedge \mathrm{d}x^k \tag{128}$$

$$\tilde{\boldsymbol{\omega}} = \hat{\omega}_\alpha \tilde{\mathbf{e}}^\alpha \tag{129}$$

where

$$\hat{\omega}_\alpha \tag{130}$$

are the Darboux tensors (antisymmetric matrices) corresponding to flow along the coordinate lines $\alpha = 1, 2, 3$ and

$$\tilde{\mathbf{e}}^\alpha \tag{131}$$

are the covector fields for the same coordinate lines.

$$\mathrm{d}Q = Q\tilde{\boldsymbol{\omega}} \tag{132}$$

$$Q \cdot e_r = \eta \tag{133}$$

10 Functional analysis and spectral methods

10.1 Green's functions

Delta function is the identity element and a Green function is the right inverse of a differential operator.

$$LG = \delta \tag{134}$$

10.2 Leray projection

Leray projection P projects a vector field on $\mathbb{R}^n$ onto its' divergence free component.

$$\begin{aligned} P(\boldsymbol{u}) &= \boldsymbol{u} - \nabla\Delta^{-1}(\nabla\cdot\boldsymbol{u}) && (135)\\ &= \frac{1}{(2\pi)^{n/2}}\int\left(\mathbf{1}-\frac{\boldsymbol{\xi}\boldsymbol{\xi}}{|\boldsymbol{\xi}|^2}\right)\cdot\hat{\mathbf{u}}(\boldsymbol{\xi})e^{i\boldsymbol{\xi}\cdot\boldsymbol{x}}\,\mathrm{d}V(\xi) && (136)\\ &= \mathscr{F}^{-1}\left(\mathbf{1}-\frac{\boldsymbol{\xi}\boldsymbol{\xi}}{|\boldsymbol{\xi}|^2}\right)\mathscr{F}\cdot\boldsymbol{u} && (137)\end{aligned}$$

In 3d the Helmholtz decomposition lets you write $\boldsymbol{u} = \nabla\phi + \nabla\times\boldsymbol{A}$ so

$$P(\boldsymbol{u}) = \nabla\times\boldsymbol{A} \tag{138}$$

11 Perturbation theory

11.1 The various sizes of 'oh'

A function ψ is...

- $o(\boldsymbol{x})$ ('little oh of $\boldsymbol{x}$') if $\lim_{\boldsymbol{x}\to 0} \frac{|\psi(\boldsymbol{x})|}{|\boldsymbol{x}|} = 0$. That is, for each $\epsilon > 0$ there is some constant $\delta > 0$ such that $|\psi(\boldsymbol{x})| \leq \epsilon|\boldsymbol{x}|$ as long as $|\boldsymbol{x}| \leq \delta$.

- $O(\boldsymbol{x})$ ('big oh of $\boldsymbol{x}$') if there is some constant c such that $|\psi(\boldsymbol{x})| \leq c|\boldsymbol{x}|$ in some neighborhood of zero.

- $f(x) = O(g(x))$ as $x \to x_*$ means there is some neighborhood V of x_* where you can find a constant $M > 0$ such that $|f(x)| \leq M|g(x)|$ for all $x \in V$.

These conditions are often written $\psi(\boldsymbol{x}) = o(\boldsymbol{x})$ or $\psi(\boldsymbol{x}) = O(\boldsymbol{x})$. Being $o(\boldsymbol{x})$ implies $O(\boldsymbol{x})$, but the converse is not true.

12 'The' Calculus

This is mostly vector/tensor calculus in 3 dimensions (i.e. the number of dimensions I actually live in.)

12.1 Differential

A function F is differentiable if it can be locally approximated by a linear map $\mathrm{d}F$ called its' differential. By 'local approximation' we mean a description of how the quantity $\boldsymbol{F}$ repsonds to changes in position (or some independent variable). I usually write 'the differential of F evaluated at the point x in the direction ξ' as $\mathrm{d}F_x(\xi)$.

$$F(x+\xi) = F(x) + \mathrm{d}F_x(\xi) + O(|\xi|^2) \tag{139}$$

$$\mathrm{d}F_x(\xi) = \frac{\partial F}{\partial x^i}\xi^i \tag{140}$$

$$\mathrm{d}F = \frac{\partial F}{\partial x^i}\,\mathrm{d}x^i \tag{141}$$

In a coordinate basis, the differential of a function is the Jacobian matrix.

$$\mathrm{d}F = \begin{bmatrix} \frac{\partial F}{\partial x^1} & \frac{\partial F}{\partial x^2} & \frac{\partial F}{\partial x^3} & \cdots \end{bmatrix} \tag{142}$$

In particular, the differential of a coordinate function x^i is just a row vector.

$$\mathrm{d}x^2 = \begin{bmatrix} 0 & 1 & 0 & \cdots \end{bmatrix} \tag{143}$$

If F is some field, then $\mathrm{d}F_x(\xi)$ is the rate at which F changes as measured by a bug passing through the point x with velocity ξ. The bug can choose any path as long

as it satisfies $\gamma(t) = x$ and $\dot{\gamma}(t) = \xi$ for some t, but bugs are lazy (and most live in flat space) so they usually pick $\gamma(t) = x + t\xi$.

$$\mathrm{d}F_x(\xi) = \frac{\mathrm{d}}{\mathrm{d}t} F(\gamma(t))\bigg|_{t=0} = \frac{\mathrm{d}}{\mathrm{d}t} F(x + t\xi)\bigg|_{t=0} \tag{144}$$

The second second differential

$$\mathrm{d}^2 f_{\boldsymbol{x}}(\xi) = \frac{\partial^2 f}{\partial x^i \partial x^j} \xi^i \xi^j \tag{145}$$

$$\mathrm{d}^2 f = \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \, \mathrm{d}x^j \tag{146}$$

The Hessian matrix is just the collection of the second differential's coefficients in a local coordinate basis.

If $\boldsymbol{F}(\boldsymbol{x})$ is a vector field it's differential can be decomposed into a sum of irreducible representations of the rotation group.

$$\mathrm{d}\boldsymbol{F} = \frac{\partial \boldsymbol{F}}{\partial x^i} \mathrm{d}x^i \tag{147}$$

$$\mathrm{d}\boldsymbol{F} = \mathrm{d}\boldsymbol{F}^{(0)} + \mathrm{d}\boldsymbol{F}^{(1)} + \mathrm{d}\boldsymbol{F}^{(2)} \tag{148}$$

- Expansion (scalar, isotropic, 1D)

$$\mathrm{d}\boldsymbol{F}^{(0)} = \frac{1}{3} \operatorname{tr}(\mathrm{d}\boldsymbol{F}) \boldsymbol{\delta} = \frac{1}{3} \nabla \cdot \boldsymbol{F} \boldsymbol{\delta} \tag{149}$$

- Rigid Rotation (vector, antisymmetric, 3D)

$$\mathrm{d}\boldsymbol{F}^{(1)} = \frac{1}{2} \left(\mathrm{d}\boldsymbol{F} - \mathrm{d}\boldsymbol{F}^T \right) = \frac{1}{2} \widehat{(\nabla \times \boldsymbol{F})} \tag{150}$$

- Pure shear (2nd rank tensor, traceless symmetric, 5D)

$$\mathrm{d}\boldsymbol{F}^{(2)} = \frac{1}{2}\left(\mathrm{d}\boldsymbol{F} + \mathrm{d}\boldsymbol{F}^T - \frac{2}{3}\,\mathrm{tr}(\mathrm{d}\boldsymbol{F})\boldsymbol{\delta}\right) \tag{151}$$

As an application, consider the problem of characterizing an isotropic background flux density (poynting vector, electric current density, momentum field of a dust particles, etc...)

$$\boldsymbol{J} : \mathbb{R}^{3+1} \to \mathbb{R}^3. \tag{152}$$

The flux density is defined, up to a constant vector field, by a gradient

$$\boldsymbol{A} = \nabla \boldsymbol{J} \tag{153}$$

which can be decomposed into irreducible components. These components represent contributions to the

$$\boldsymbol{A}^{(0)} = \frac{1}{3}\left(\nabla \cdot \boldsymbol{J}\right)\boldsymbol{\delta} \tag{154}$$

$$\boldsymbol{A}^{(1)} = \frac{1}{2}\left(\nabla_a J_b - \nabla_b J_a\right)\boldsymbol{E}^a \boldsymbol{E}^b = \frac{1}{2}\varepsilon_{klm}\left(\nabla \times \boldsymbol{J}\right)^l \boldsymbol{E}^k \boldsymbol{E}^m = \frac{1}{2}\widehat{(\nabla \times \boldsymbol{J})} \tag{155}$$

$$\boldsymbol{A}^{(1)} = \frac{1}{2}\left(\nabla_a J_b - \nabla_b J_a\right)\boldsymbol{E}^a \boldsymbol{E}^b = \frac{1}{2}\varepsilon_{klm}\left(\nabla \times \boldsymbol{J}\right)^l \boldsymbol{E}^k \boldsymbol{E}^m = \frac{1}{2}\widehat{(\nabla \times \boldsymbol{J})} \tag{156}$$

12.2 The Del operator ∇

Like the exterior derivative (differential) d, the Del operator ∇ provides a description of changes to a function or field due to changes changes in position. The divergence and curl of vector calculus are essentially the first two irreducible components of

the differential. **Curl**

In curvilinear coordinates

$$\nabla \times \boldsymbol{F} = \boldsymbol{e}_i \epsilon^{ijk} \nabla_j F_k \tag{157}$$

$$(\nabla \times \boldsymbol{F})^i = \epsilon^{ijk} \nabla_j F_k \tag{158}$$

Theorem : If $\nabla g(\boldsymbol{x}) \neq 0$, then it is normal to level sets of the function g. More precisely, if $S = \left\{ \boldsymbol{z} \in \mathbb{R}^3 \middle| g(\boldsymbol{z} = \xi \right\}$ for some $\xi \in \mathbb{R}$ is a level$\boldsymbol{v} \in T_{\boldsymbol{x}} g^{-1}(\boldsymbol{x})$

$$\boldsymbol{v} \cdot \nabla g(\boldsymbol{x}) \qquad \forall \boldsymbol{v} \in T_{\boldsymbol{x}} S \tag{159}$$

Proof: Any curve through the surface satisfies $g(\boldsymbol{x}(t)) = c$, so, $\frac{\mathrm{d}}{\mathrm{d}t} g(\boldsymbol{x}(t)) = \dot{\boldsymbol{v}} \cdot \nabla g = 0$.

□

Theorem : If $\nabla g(\boldsymbol{x}_0) \neq 0$, then each level sets of $g(\boldsymbol{x})$ which containts $\boldsymbol{x}_0$ is a smooth surface near $\boldsymbol{x}_0$.

Proof: implicit function theorem □

The laplacian and angular momentum operator

The position space representation of the angular momentum operator is[1]

$$\boldsymbol{L} = -i\boldsymbol{x} \times \nabla = -i\boldsymbol{E}_k \varepsilon_{ijk} x_i \partial_j = -i\left(\boldsymbol{e}_\phi \frac{\partial}{\partial \theta} - \boldsymbol{e}_\theta \frac{1}{\sin\theta} \frac{\partial}{\partial \phi} \right). \tag{160}$$

Dotting with a vector $\boldsymbol{n}$ gives you the directional derivative in the $\boldsymbol{n} \times \boldsymbol{x}$-direction. Note that it's scaled by $\boldsymbol{x}$ so points farther from the origin are weighted more heavily.

$$\boldsymbol{n} \cdot (\boldsymbol{x} \times \nabla) = (\boldsymbol{n} \times \boldsymbol{x}) \cdot \nabla \tag{161}$$

[1] In units with $\hbar = 1$. Otherwise you need to multiply by $\hbar$.

The total squared angular momentum

$$-\boldsymbol{L}^2 f = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}. \tag{162}$$

is essentially the spherical part of the Laplacian

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{1}{r^2}\boldsymbol{L}^2 \tag{163}$$

$$\nabla = \boldsymbol{e}^k \nabla_k \tag{164}$$

$$\delta\boldsymbol{x} = \boldsymbol{e}_k \delta x^k \tag{165}$$

$$\boldsymbol{L} = -i\hbar\boldsymbol{x}\times\nabla = -i\hbar\boldsymbol{e}^i \varepsilon_{ij.}^{\cdot\cdot k} x^j \nabla_k \tag{166}$$

$$\delta\hat{\mathbf{x}} \tag{167}$$

12.3 Integration

Theorem Change of variables: Let $\xi = \Gamma(x)$ define a coordinate transformation on some nice region $U \subset \mathbb{R}^n$ (so, x's live in U and ξ's live in $\Gamma(U)$).

$$\int_U f(\boldsymbol{x})\,\mathrm{d}^n x = \int_{\Gamma(U)} f(\boldsymbol{x}(\boldsymbol{\xi}))\left|\det\frac{\partial x}{\partial\xi}\right|^{-1}\mathrm{d}^n\xi \tag{168}$$

From Termodynamics and Thermo-statistics by Callen:

$$\frac{\partial(u,v,\ldots,w)}{\partial(x,y,\ldots,z)} := \det \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \cdots & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \cdots & \frac{\partial v}{\partial z} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \cdots & \frac{\partial w}{\partial z} \end{bmatrix} \tag{169}$$

$$\frac{\partial(u,v,\ldots,w)}{\partial(x,y,\ldots,z)} = -\frac{\partial(v,u,\ldots,w)}{\partial(x,y,\ldots,z)} \tag{170}$$

$$\frac{\partial(u,v,\ldots,w)}{\partial(x,y,\ldots,z)} = \frac{\partial(u,v,\ldots,w)}{\partial(r,s,\ldots,t)} \frac{\partial(r,s,\ldots,t)}{\partial(x,y,\ldots,z)} \tag{171}$$

$$\frac{\partial(x,y,\ldots,z)}{\partial(u,v,\ldots,w)} = \frac{1}{\frac{\partial(u,v,\ldots,w)}{\partial(x,y,\ldots,z)}} \tag{172}$$

$$\left(\frac{\partial u}{\partial x}\right)_{y,\ldots,z} := \frac{\partial(u,y,\ldots,z)}{\partial(x,y,\ldots,z)} \tag{173}$$

A valid set of coordinate functions is nothing more than

12.4 Invariant integration

A region V in some physical/geometric 'space' exists independent of any particular parametrization. Similarly, physical quantities or fields exist regardless whether or not we have chosen coordinates to describe them. One would imagine that there is a way to describe things like volumes and integrals of other physical quantities in a coordinate independent, or, at the very least, in a covariant[2] manner. Well, it turns

[2] i.e. formulas have the same form in all coordinate systems

out you can. Big surprise, I know.

position vector: $\boldsymbol{r}$

covariant basis: $\boldsymbol{g}_i = \frac{\partial \boldsymbol{r}}{\partial \xi^i}$

metric: $g_{ij}(\xi)$

volume element: $\mathrm{d}V(\xi) = \boldsymbol{g}_3 \cdot \left(\boldsymbol{g}_1 \times \boldsymbol{g}_2\right) = \sqrt{\left|\det g(\xi)\right|}\,\mathrm{d}^n\xi$

Invariant integral:

$$\int_V F\,\mathrm{d}V = \int_{V(x)} F(x)\,\mathrm{d}V(x) = \int_{V(\xi)} F(\xi)\,\mathrm{d}V(\xi) \tag{174}$$

We are thinking of F as a field rather than a function in the normal calculus sense. This means that the quantity $F(\xi)$ should be interpreted as the value of the field F at the *physical point* (say p) whose coordinate in the chosen coordinate system is ξ. In a different coordinate system where p has coordinate x we would just write $F(x)$ for the same physical quantity. In contrast, if we were treating F as a 'normal' function we would need to write something like $F(\xi(x))$.

This is all consistent with the normal analysis/calculus change of variables formula! If $\boldsymbol{J}(x)$ is the Jacobian matrix of the coordinate transformation $\xi = \xi(x)$, then

$$\boldsymbol{J}^{-1}(\xi) = \begin{bmatrix} \frac{\partial x^1}{\partial \xi^1} & \frac{\partial x^2}{\partial \xi^1} & \cdots \\ \frac{\partial x^1}{\partial \xi^2} & \frac{\partial x^2}{\partial \xi^2} & \cdots \\ \vdots & & \ddots \end{bmatrix}, \tag{175}$$

and since $g_{ij}(\xi) = g_{kl}(x)\frac{\partial x^k}{\partial \xi^i}\frac{\partial x^l}{\partial \xi^j}$ we can write

$$\boldsymbol{g}(\xi) = \boldsymbol{J}^{-1}(\xi) \cdot \boldsymbol{g}(x(\xi)) \cdot \boldsymbol{J}^{-T}(\xi) \tag{176}$$

$$\det\big(g(\xi)\big) = \frac{1}{\det(J)^2}\det\big(g(x)\big) \tag{177}$$

$$\mathrm{d}^n x = \frac{\sqrt{\left|\det\big(g(\xi)\big)\right|}}{\sqrt{\left|\det\big(g(x)\big)\right|}}\,\mathrm{d}^n\xi = \frac{1}{|\det(J(x))|}\,\mathrm{d}^n\xi \tag{178}$$

invariant delta functions

The density for a unit point mass at ξ' is

$$\delta_{\xi'}(\xi) = \frac{1}{\sqrt{\left|\det g(\xi)\right|}}\delta(\xi-\xi') \tag{179}$$

where $\delta(\xi-\xi') = \delta(\xi^1-\xi'^1)\delta(\xi^2-\xi'^2)\ldots$ is the normal delta function. In a cartesian coordinates this reduces to the $\delta_{x'} = \delta(x-x')$.

Theorem Green's 2nd Identity:

$$\int_V \left[f\nabla^2 g - g\nabla^2 f\right]\mathrm{d}V = \int_{\partial V}\left[f\nabla g - g\nabla f\right]\cdot\mathrm{d}\boldsymbol{S} \tag{180}$$

12.5 Fourier stuff

Fourier transform:

$$\hat{u}(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} u(x)e^{-ikx}\,\mathrm{d}x \tag{181}$$

$$u(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \hat{u}(k)e^{ikx}\,\mathrm{d}k \tag{182}$$

Fourier series

For a function $f(x)$ defined on $-\pi < x \leq \pi$, define $\tilde{f}(x)$ to be its 2π-periodic extension.

$$\tilde{f}(x+2\pi m) = f(x), \qquad \text{for some } m(x) \in \mathbb{Z} \tag{183}$$

$$\tilde{f}(\pi) = \tilde{f}(-\pi) = \frac{f(\pi)+f(-\pi)}{2} \tag{184}$$

The Fourier series of $f(x)$ is

$$f(x) \sim a_0/2 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) \tag{185}$$

$$f(x) \sim \sum_{k=-\infty}^{\infty} c_k e^{ikx} \tag{186}$$

Theorem : If $f(x) \sim \sum_k c_k e^{ikx}$ with $\sum_k |c_k| < \infty$ the the FS cxonverges uniformly to continuous $\tilde{f}(x)$.

Theorem : Let $f(x)$ be 2π-periodic and pw C^1. If f is continuous on some $a < x < b$ then its' FS converges uniformly to f on any closed interval inside of $a < x < b$.

Theorem : If f is piecewise-continuous and is mean zero ($a_0 = 0$) on $-\pi < x \leq \pi$ with

$$f(x) \sim \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx), \tag{187}$$

then

$$g(x) := \int_0^x f(y)\,\mathrm{d}y \sim m + \sum_{k=1}^{\infty} \frac{-b_k}{k}\cos(kx) + \frac{a_k}{k}\sin(kx) \tag{188}$$

where $m = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(x)\,\mathrm{d}x$.

Theorem : If f has pw-C^2 and continuous 2π periodic extension then

$$f'(x) \sim \sum_{k=1}^{\infty} kb_k \cos(kx) - ka_k \sin(kx) = \sum_{k=-\infty}^{\infty} ikc_k e^{ikx} \tag{189}$$

Theorem : Let $n \in \mathbb{Z}$ $(n \geq 0)$, $f(x) \sim \sum_k c_k e^{ikx}$, and $\sum_k |k^n c_k| < \infty$. Then the FS converges uniformly to $\tilde{f} \in C^n$ and the m-times differentiated FS converges uniformly to $\frac{\mathrm{d}^m \tilde{f}}{\mathrm{d}x^m}$.

Theorem : Let g be a continuously differentiable map from some open subset of a linear space X to a linear space Y. If $\mathrm{d}g_x$ is onto at every point in the $S_u = \{x \in X | g(x) = u\}$, then S_u is an n-dimensional submanifold of X.

Proof: This is thm 12.1 from advanced calculus loomis, sternberg □

Say you want to minimize $f(x)$ subject to constraints $g_i(x) = u_i$. Define the Lagrangian function $L(x, \lambda) = f(x) - \sum_i \lambda_i \left(g_i(x) - u_i\right)$

Fourier transform is a (unitary wrt $\|\cdot\|_{L^2}$) operator on functions/distributions

$$\mathscr{F} f \tag{190}$$

Convolution is a binary operation on functions/distributions

$$(f * g)(x) = \int f(x')g(x - x')\,\mathrm{d}x' \tag{191}$$

12.6 Lagrange multipliers

Say you want to minimize f subject to $g(x) = c$. As long a g is nice, the constraint defines a submanifold M. The projection of ∇f onto the tangent space of M vanishes at critical points (in M), so it must be normal to the M. Since ∇g is also normal to M, it must be proportional to ∇f.

$$\nabla f = \lambda \nabla g \tag{192}$$

12.7 Some boring theorems, definitions, technical stuff, etc...

Theorem : Term by term differentiation works for power series inside range of convergence

Definition ():$v_n(x)$ converges uniformly to $v(x)$ on $I \subset \mathbb{R}$ if for every $\epsilon > 0$ there is some N with $|v_n(x) - v(x)| < \epsilon$ for all $x \in I$ whenever $n > N$.

Theorem : If $v_n(x)$'s are continuous and $v_n \to v$ uniformly, then v is continuous.

Theorem : If $|u_k(x)| \leq m_k$ for all $x \in I$ and $\sum_k m_k < \infty$ then $\sum_k u_k(x)$ converges uniformly.

Theorem : If $\sum_k u_k'(x) = g(x)$ is uniformly convergent then so is $\sum_k u_k(x) = f(x)$, and $f'(x) = g(x)$.

- If all partial derivatives of a function exist in a neighborhood of $\boldsymbol{x}$ and are continuous at $\boldsymbol{x}$ then f is differentiable at $\boldsymbol{x}$.

Inverse function theorem:

Theorem : If $b = f(a)$ then $\left(f^{-1}\right)'(b) = \frac{1}{f'(a)}$ wherever all that stuff is defined.

12.8 Rotations and other Lie group stuff

I think maybe this is from **Jeff's mechanics Lecture notes:**

If $\boldsymbol{e}_i, \boldsymbol{d}_i$ are orthonormal bases (with the same orientation) for $\mathbb{R}^3$ they're related by a rotation

$$\boldsymbol{d}_i = R_{ij}\boldsymbol{e}_j. \tag{193}$$

$$\boldsymbol{d}_i \cdot \boldsymbol{e}_j = R_{ij} =: \cos\theta_{ij} \tag{194}$$

It kind of even makes sense to just define rotations to be transformations that map orthonormal bases to each other.

You can specify a rotation by direction cosines α, β, γ of its' rotation axis $\boldsymbol{n} = (\cos\alpha, \cos\beta, \cos\gamma)$ and an angle of rotation θ.[3]

$$R_{\boldsymbol{n}}(\theta) = \cos\frac{\theta}{2} - \boldsymbol{\sigma}\cdot\boldsymbol{n}\sin\frac{\theta}{2} = \cos\frac{\theta}{2} - \left(i\cos\alpha + j\cos\beta + k\cos\gamma\right)\sin\frac{\theta}{2} \tag{195}$$

This is a 2-to-1 correspondence since

$$R_{\boldsymbol{n}}(\theta) = R_{-\boldsymbol{n}}(-\theta) \tag{196}$$

I guess dotting with σ is essentially like the hat map... Vectors are like pure quaternions and rotations are like quaternions with a real part.

$$\boldsymbol{r}'\cdot\boldsymbol{\sigma} = R^{\dagger}_{\boldsymbol{n}}(\theta)\,(\boldsymbol{r}\cdot\boldsymbol{\sigma})\,R_{\boldsymbol{n}}(\theta) \tag{197}$$

Note that $R^{\dagger}_{\boldsymbol{n}}(\theta) = R_{\boldsymbol{n}}(-\theta)$. As a matrix (Rodrigues formula)

$$R_{\boldsymbol{n}}(\theta) = \boldsymbol{nn} + \cos\theta\,(\boldsymbol{\delta} - \boldsymbol{nn}) + \sin\theta\hat{\mathbf{n}} \tag{198}$$

$$\mathrm{tr}(R_{\boldsymbol{n}}(\theta)) = 1 + 2\cos\theta \tag{199}$$

From **Lie-group methods by iserles** (check eq 41 and 6.16):
Consider the matrix ODE

$$Y' = A(t)Y, \qquad t \geq 0, \qquad Y(0) = Y_0. \tag{200}$$

If Y_0 is in a Lie group G and A is in its' Lie algebra $\mathfrak{g}$, then $Y(t)$ stays in G. For a

[3] check the minus sign...

rotation, if $\boldsymbol{c}_i$ is a basis of $\mathbb{R}^3$ then there are scalar functions $\theta_i(t)$ such that

$$Y(t) = e^{\theta_1(t)\hat{\mathbf{c}}_1} e^{\theta_2(t)\hat{\mathbf{c}}_2} e^{\theta_3(t)\hat{\mathbf{c}}_3} \tag{201}$$

$$\hat{\mathbf{A}}(0) = \sum_i \theta_i'(0)\hat{\mathbf{c}}_k \tag{202}$$

12.9 Curvilinear Coordinates

- coordinates z^i
- Position vector $\boldsymbol{x} = \boldsymbol{x}(z)$
- covariant basis $\boldsymbol{z}_i = \frac{\partial \boldsymbol{x}(z)}{\partial z^i}$
- covariant metric components $z_{ij} = \boldsymbol{z}_i \cdot \boldsymbol{z}_j$
- contravariant metric components $\left[z^{ij}\right] = \left[z_{ij}\right]^{-1}$
- covariant derivative and the gradient $\nabla \boldsymbol{v} = \boldsymbol{z}^i \nabla_i$

$$\nabla\left(v^a \boldsymbol{e}_a\right) = \nabla\left(v^a\right)\boldsymbol{e}_a = \frac{\partial v^a}{\partial x^b}\boldsymbol{e}^b \boldsymbol{e}_a = \frac{\partial v^a}{\partial z^i}\frac{\partial z^i}{\partial x^b}\boldsymbol{e}^b \boldsymbol{e}_a = \frac{\partial v^a}{\partial z^i}\boldsymbol{z}^i \boldsymbol{e}_a \tag{203}$$

$\frac{\partial z^i}{\partial x^b}\boldsymbol{e}^b \cdot \boldsymbol{z}_j = \frac{\partial z^i}{\partial x^b}\boldsymbol{e}^b \cdot \frac{\partial \boldsymbol{x}}{\partial z_j} = \frac{\partial z^i}{\partial x^b}\frac{\partial x^b}{\partial z_j} = \frac{\partial z^i}{\partial z_j} = \delta^i_j$ so $\frac{\partial z^i}{\partial x^b}\boldsymbol{e}^b = \boldsymbol{z}^i$.

13 Constrained optimization

13.1 Notes from Marsden

Let V and Λ be Banach spaces and $\phi : V \to \Lambda$ a smooth map. Suppose y is a regular value of ϕ so that $C := \phi^{-1}(y)$ is a smooth submanifold of V. Let $h : V- \to \mathbb{R}$ be smooth and define $\overline{h} : V \times \Lambda^* \to \mathbb{R}$ by

$$\overline{h}(x, \lambda) = h(x) - \langle \lambda, \phi(x) \rangle \tag{204}$$

Theorem : The following are equivalent

- x_0 is a critical point of $h|_C$
- there is some $\lambda_0 \in \Lambda^*$ such that (x_0, λ_0) is a critical point of $\overline{h}$

Two submanifolds of a given finite-dimensional smooth manifold are said to intersect transversally if at every point of intersection, their separate tangent spaces at that point together generate the tangent space of the ambient manifold at that point.

Definition (Differentiable): A function $F : V \to W$ between normed linear spaces is differentiable at $\alpha \in V$ if there is linear map $\mathrm{d}F_\alpha : V \to W$ (called the differential of F at α) which satisfies $F(\alpha + \xi) - F(\alpha) = \mathrm{d}F_\alpha(\xi) + o(\xi)$

1. Calculus of variations texts like to call this the first variation, but they really are the same thing.
2. $\mathrm{d}(F + G)_\alpha = \mathrm{d}F_\alpha + \mathrm{d}G_\alpha$
3. $F \in \mathscr{D}_\alpha(V, \mathbb{R})$ and $G \in \mathscr{D}_\alpha(V, W)$ then[4] $\mathrm{d}(FG)_\alpha = F(\alpha)\,\mathrm{d}G_\alpha + \mathrm{d}F_\alpha\, G(\alpha)$ where $\mathrm{d}(FG)_\alpha(\xi) = F(\alpha)\,\mathrm{d}G_\alpha(\xi) + \mathrm{d}F_\alpha(\xi) G(\alpha)$
4. If F is constant on V, then $\mathrm{d}F_\alpha = 0$

[4] $FG \in \mathscr{D}_\alpha(V, W)$ is the map $\alpha \mapsto F(\alpha)G(\alpha)$

5. If $F \in Hom(V, W)$, then F is differentiable at every $\alpha \in V$ and $\mathrm{d}F_\alpha = F$. (e.g. the identity map, coordinate functions, or any projection operator)

6. If F is differentiable at α and γ is any smooth curve through α with

Theorem : If $F \in \mathscr{D}_\alpha(V, W)$ and $G \in \mathscr{D}_{F(\alpha)}(W, X)$, then $G \circ F \in \mathscr{D}_\alpha(V, X)$ and $\mathrm{d}(G \circ F)_\alpha = \mathrm{d}G_{F(\alpha)} \circ \mathrm{d}F_\alpha$

If γ is a smooth parameterized curve through $\alpha = \gamma(t)$ then its tangent vector at α is $\gamma'(t) = \lim_{s \to 0} \frac{\gamma(t+s) - \gamma(t)}{s}$. Since we know the differential is a linear map we can write $\frac{\gamma(t+s) - \gamma(t)}{s} = \mathrm{d}\gamma(1) + o(s)/s$ and hence $\gamma'(t) = \mathrm{d}\gamma(1)$. Also, γ is differentiable at t iff its tangent vector exists.

Definition (): Let F be a function from V to W and let γ be a smooth parameterized curve through $\alpha = \gamma(t)$ with tangent vector $\gamma'(t) = \xi$. The directional derivative of F in the direction of ξ at α, denoted $D_\xi F(\alpha)$, is just the tangent vector to the parametrized curve $F \circ \gamma$ at $F(\alpha) = F(\gamma(t))$.

By above stuff, $D_\xi F(\alpha) = \frac{\mathrm{d}}{\mathrm{d}t} F(\gamma(t)) = \mathrm{d}(F \circ \gamma)_t(1) = \mathrm{d}F_{\gamma(t)}(\mathrm{d}\gamma_t(1)) = \mathrm{d}F_{\gamma(t)}(\gamma'(t)) = \mathrm{d}F_\alpha(\xi)$.

Once a basis ∂_i is chosen, partial derivatives can just be defined as directional derivatives in the direction of the basis vectors $\frac{\partial F}{\partial x^i} := \mathrm{d}F_\alpha(\partial_i)$.

A bounded bilinear mapping $\omega : X \times Y \to W$, that is $\|\omega(\xi, \eta)\| \leq b\|\xi\|\|\eta\|$ for some constant b, is everywhere differentiable and $\mathrm{d}\omega_{(\alpha,\beta)}(\xi, \eta) = \omega(\alpha, \eta) + \omega(\xi, \beta)$.

Theorem : Let F be a real valued function defined on an open neighborhood A of $\alpha \in V$ and suppose that F attains a local max/min at α where $\mathrm{d}F_\alpha$ exists. Then $\mathrm{d}F_\alpha = 0$.

14 Differential equations

14.1 Stokes Equation

$$\begin{aligned} \nabla^2 \boldsymbol{u} &= -\frac{1}{\mu}\nabla p \\ \nabla \cdot \boldsymbol{u} &= 0 \end{aligned} \tag{205}$$

14.2 Poisson equation

(associated) legendre, bessel, spherical harmonics +addition theorem

14.3 Poisson equation

$$\nabla^2 u(\boldsymbol{x}) = -\rho(\boldsymbol{x}) \tag{206}$$

The Green's function for Eq (206) is

$$G(\boldsymbol{x}-\boldsymbol{x}') = \frac{1}{4\pi}\frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|} + F(\boldsymbol{x},\boldsymbol{x}'), \tag{207}$$

where F is a harmonic function chosen to satisfy boundary conditions,

$$\nabla^2 G(\boldsymbol{x}-\boldsymbol{x}') = -\delta(\boldsymbol{x}-\boldsymbol{x}'), \qquad \nabla^2 F(\boldsymbol{x},\boldsymbol{x}') = 0. \tag{208}$$

$$u(\boldsymbol{x}) = \int_V \rho(\boldsymbol{x}')G(\boldsymbol{x}-\boldsymbol{x}')\,\mathrm{d}V(\boldsymbol{x}') + \int_{\partial V}\left[G(\boldsymbol{x}-\boldsymbol{x}')\nabla' u(\boldsymbol{x}') - \nabla' G(\boldsymbol{x}-\boldsymbol{x}')u(\boldsymbol{x}')\right]\cdot \mathrm{d}\boldsymbol{S}(\boldsymbol{x}') \tag{209}$$

14.4 Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u \tag{210}$$

1D Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{211}$$

d'Lambert's Solution

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \qquad u(0,x) = f(x), \qquad \frac{\partial u}{\partial t} = g(x), \qquad -\infty < x < \infty \tag{212}$$

$$u(t,x) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(z)\,\mathrm{d}z. \tag{213}$$

Wave equation with source

(I think this is all from Jackson)

$$\nabla^2 \psi - \frac{1}{c^2}\frac{\partial^2 \psi}{\partial t^2} = -4\pi f(\boldsymbol{x}, t) \tag{214}$$

$$\psi(\boldsymbol{x}, t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{\psi}(\boldsymbol{x}, \omega) e^{-i\omega t}\,\mathrm{d}\omega, \qquad \tilde{\psi}(\boldsymbol{x}, \omega) = \int_{-\infty}^{\infty} \psi(\boldsymbol{x}, t) e^{i\omega t}\,\mathrm{d}t \tag{215}$$

$$\nabla^2 \tilde{\psi} + k^2 \tilde{\psi} = -4\pi \tilde{f} \tag{216}$$

where $k^2 = (\omega/c)^2$.

$$G_k^{(\pm)}(\boldsymbol{x}, t; \boldsymbol{x}', t') = \frac{\delta(t - t' \pm |\boldsymbol{x} - \boldsymbol{x}'|/c)}{|\boldsymbol{x} - \boldsymbol{x}'|} \tag{217}$$

$$\left[\nabla^2 + k^2\right] G_k^{(\pm)}(\boldsymbol{x}, t; \boldsymbol{x}', t') = -4\pi\delta(\boldsymbol{x} - \boldsymbol{x}') \tag{218}$$

By symmetry, only depends on $|\boldsymbol{x} - \boldsymbol{x}'| = R$

$$\frac{1}{R}\frac{\mathrm{d}^2}{\mathrm{d}R^2}\left[R^2 G_k\right] + k^2 G_k = -4\pi\delta(R) \tag{219}$$

$$\left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right] G_k^{(\pm)}(\boldsymbol{x}, t; \boldsymbol{x}', t') = -4\pi\delta(\boldsymbol{x} - \boldsymbol{x}')\delta(t - t') \tag{220}$$

$$\psi(\boldsymbol{x}, t) = \int \frac{\left[f(\boldsymbol{x}', t')\right]_{ret}}{|\boldsymbol{x} - \boldsymbol{x}'|}\,\mathrm{d}^3 x' \tag{221}$$

where $t' = t - |\boldsymbol{x} - \boldsymbol{x}'|/c$.

14.5 Screened Poisson equation

$$\left[\nabla^2 - \lambda^2\right] u(\boldsymbol{x}) = \rho(\boldsymbol{x}) \tag{222}$$

14.6 Navier-Stokes equations

14.6.1 Stokes equations (Low Reynolds number)

$$-\nabla p(\boldsymbol{x} + \eta \nabla^2 \boldsymbol{v}(\boldsymbol{x}) + \boldsymbol{f}(\boldsymbol{x}) = 0 \tag{223}$$

$$\nabla \boldsymbol{v}(\boldsymbol{x}) = 0 \tag{224}$$

14.7 Sturm-Liouville theory

Given functions $p(x), q(x)$ and *weight* $w(x) > 0$, find eigenvalue λ and eigenfunction $u(x)$ satisfying

$$(p(x)u'(x))' + q(x)u(x) + \lambda w(x)u(x) = 0 \tag{225}$$

subject to some boundary conditions (e.g. $u(a) = 0 = u(b)$). If above things satisfy: $p(x), w(x) > 0$; p, p', q, w are continuous on $[a, b]$; boundary conditions are given by

$$\alpha_1 u(a) + \alpha_2 u'(a) = 0 \qquad \alpha_1^2 + \alpha_2^2 > 0 \tag{226}$$

$$\beta_1 u(b) + \beta_2 u'(b) = 0 \qquad \beta_1^2 + \beta_2^2 > 0 \tag{227}$$

the problem is said to be regular and we can say some nifty stuff.

- eigenvalues λ_n are countable, can be indexed so as to be strictly increasing $\lambda_n < \lambda_{n+1}$, and are unbounded $\lambda_n \to \infty$
- corresponding (normalized)eigenfunctions $u_n(x)$ have exactly $n-1$ zeros in $[a, b]$ and are orthonormal wrt the weighted inner product

$$\int_a^b u_m(x)u_n(x)w(x)\,\mathrm{d}x = \delta_{mn} \tag{228}$$

Define the differential operator $Lu = (pu')' + qu$ so s-l problem is

$$\big(Lu\big)(x) = -\lambda w(x)u(x). \tag{229}$$

and if u, v satisfy the regular bc's above we have

$$\int_a^b vLu\,\mathrm{d}x = \int_a^b (Lv)u\,\mathrm{d}x. \tag{230}$$

Equivalent variational problem: define

$$I[u] = \frac{\int_a^b p(x)(u'(x))^2 - q(x)u(x)^2\,\mathrm{d}x}{\int_a^b u(x)^2 w(x)\,\mathrm{d}x} \tag{231}$$

If u_0 minimizes $I[u]$

$$\delta I[u_0, \delta u] = 0 \qquad \forall \delta u \text{ with } \delta u(a) = \delta u(b) = 0 \tag{232}$$

then u_0 solves the eigenvalue problem with $\lambda = I[u_0]$

$$\big(Lu_0\big)(x) = -I[u_0]\,w(x)\,u(x) \tag{233}$$

14.8 Associated Laguerre equation

For $\lambda, \nu \in \mathbb{R}$

$$xy'' + (\nu + 1 - x)y' + \lambda y = 0 \tag{234}$$

Solutions $U(-\lambda, 1+\nu, x)$ where U is a confluent hypergeometric function of the first kind and when $\lambda = n = 0, 1, 2, 3, \ldots$ associated Laguerre polynomials $L_n^\nu(x)$.

- comes up in solutions to schrodinger equation. U solutions for quantum defect or something. something something $1/r$ potentials...

14.8.1 Laplace's Equation, Associated Legendre Polynomials, and Spherical Harmonics

Laplace's equation

$$\nabla^2 \Phi = 0 \tag{235}$$

describes in a region where $\rho = 0$. In spherical coordinates, the general solution (obtained by separation of variables) is given by

$$\Phi(\boldsymbol{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \left[A_{lm} r^{l} + B_{lm} r^{-(l+1)} \right] Y_{lm}(\theta, \phi) \tag{236}$$

where A_{lm} and B_{lm} are chose to satisfy boundary conditions and Y_{lm} are spherical harmonics.

14.9 Symmetries of differential equations and Lie groups

15 Special Functions

special functions course

15.1 Chebyshev Polynomials

15.2 Hermite Polynomials

Generating function:

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{t^n}{n!} H_n(x) \tag{237}$$

Some identities:

$$H_n(x) = (-1)^n e^{x^2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2} \tag{238}$$

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{1}{n!} x^n H_n(x) \tag{239}$$

$$H_n'(x) = 2nH_{n-1}(x) \tag{240}$$

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \tag{241}$$

$$\int_{-\infty}^{\infty} e^{-x^2} H_n(x) H_m(x)\,\mathrm{d}x = \sqrt{\pi} 2^2 n! \delta_{nm} \tag{242}$$

15.3 Associated Legendre polynomials

$$(1-x^2)P_l'^m(x) - 2xP_l'^m(x) + (l(l+1) - \tfrac{m^2}{1-x^2})P_l^m(x) = 0 \tag{243}$$

$$P_l^m(x) = (-1)^m(1-x^2)^{m/2}D^m P_l(x) \tag{244}$$

$$= \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}D^{l+m}(x^2-1)^l \tag{245}$$

- alternate convention $P_{lm} = (-1)^m P_l^m$

For each fixed m, the P_l^m's are *orthogonal* on $[-1,1]$.

$$\int_{-1}^{1} P_{l'}^m(x)P_l^m(x)\,\mathrm{d}x = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{l'l} \tag{246}$$

$$P_l^{-m} = (-1)^m\frac{(l-m)!}{(l+m)!}P_l^m \tag{247}$$

15.4 Spherical harmonics

$$Y_{lm}(\theta,\phi) = \sqrt{\tfrac{(2l+1)(l-m)!}{4\pi(l+m)!}}P_l^m(\cos\theta)e^{im\phi} \tag{248}$$

$$= e^{im\phi}\sum_{k=\max(0,-m)}^{\left\lfloor\frac{l-m}{2}\right\rfloor} N_{lmk}\cos^{l-(m+2k)}(\theta)\sin^{m+2k}(\theta) \tag{249}$$

$$N_{lmk} = \sqrt{\frac{(sl+1)(l+m)!(l-m)!}{}} \tag{250}$$

- $l = 0,1,2,\ldots$ and $m = -l,-l+1,\ldots,l-1,l$
- $P_l^m(x)$ are associated Legendre polynomials.
- Sometimes written Y_l^m (math literature)

$$\int_S Y^*_{l'm'}(\theta,\phi)\,Y_{lm}(\theta,\phi)\,\mathrm{d}\Omega = \delta_{l'l}\delta_{m'm} \tag{251}$$

$$\nabla^2_S Y_{lm}(\theta,\phi) = -l(l+1)\,Y_{lm}(\theta,\phi) \tag{252}$$

$$\sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y^*_{lm}(\theta',\phi')\,Y_{lm}(\theta,\phi) = \delta(\phi-\phi')\delta(\cos\theta-\cos\theta') \tag{253}$$

$$Y_{l,-m} = (-1)^{m+l}\,Y^*_{lm} \tag{254}$$

$$Y_{lm}(-\boldsymbol{n}) = (-1)^l\,Y_{lm}(\boldsymbol{n}) \tag{255}$$

$$\frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|} = \sum_{l=0}^{\infty}\frac{4\pi}{2l+1}\frac{r_<^l}{r_>^{l+1}}\sum_{m=-l}^{m=l} Y^*_{lm}(\theta',\phi')\,Y_{lm}(\theta,\phi) \tag{256}$$

- Orthonormal on S^2

$$\int_S Y^*_{l'm'}(\theta,\phi)\,Y_{lm}(\theta,\phi)\,\mathrm{d}\Omega = \delta_{m'm}\frac{(2l+1)(l-m)!}{2(l+m)!}\int_{-1}^{1} P_{l'm}(x)P_{lm}(x)\,\mathrm{d}x \tag{257}$$

$$= \delta_{l'l}\delta_{m'm} \tag{258}$$

- Eigenfunctions of spherical laplacian (angular momentum op).

$$\nabla^2_S Y_{lm}(\theta,\phi) = N_{lm}e^{im\phi}\Big[\sin^2\theta P''_{lm}(\cos\theta) - 2\cos\theta P'_{lm}(\cos\theta) - \tfrac{m^2}{\sin^2\theta}P_{lm}(\cos\theta)\Big] \quad (\text{ let } x=\cos\theta) \tag{259}$$

$$= N_{lm}e^{im\phi}\Big[(1-x^2)\theta P''_{lm}(x) - 2xP'_{lm}(x) - \tfrac{m^2}{1-x^2}P_{lm}(x)\Big] \tag{260}$$

$$= N_{lm}e^{im\phi}\Big[-l(l+1)P_{lm}(x)\Big] \tag{261}$$

$$= -l(l+1)\,Y_{lm}(\theta,\phi) \tag{262}$$

Where $N_{lm} = \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}}$

$$r^l Y_{lm} = A_{lm} \sum_{p,q,s=1}^{\infty} \frac{1}{p!q!s!} \left(-\frac{x+iy}{2}\right)^p \left(\frac{x-iy}{2}\right)^q z^s \delta_{p+q+s,l} \delta_{p-q,m} \tag{263}$$

$$A_l m = \sqrt{(2l+1)(l+m)!(l-m)!/(4\pi)} \tag{264}$$

$$\boldsymbol{e}_\pm = \mp \frac{1}{\sqrt{2}} (\boldsymbol{e}_x \pm i \boldsymbol{e}_y) \tag{265}$$

$$\boldsymbol{e}_0 = \boldsymbol{e}_z \tag{266}$$

$$\boldsymbol{e}_r = \sin\theta\cos\phi \boldsymbol{e}_x + \sin\theta\sin\phi \boldsymbol{e}_y + \cos\theta \boldsymbol{e}_z \tag{267}$$

$$= ...?Y_{1m} \boldsymbol{e}_m \tag{268}$$

15.5 Bessel functions

Bessel functions of the first kind of integer order, $J_n(x)$ are solutions to the Bessel DE

$$x^2 y'' + x y' + (x^2 - n^2) y = 0 \tag{269}$$

$$e^{\frac{x}{2}(t-1/t)} = \sum_{m=0}^{\infty} J_m(x) t^m \tag{270}$$

I have a proof of this somewhere in either an EM or dynamics hw.

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} m!(m+n)!} \tag{271}$$

$$J_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(n\tau - x\sin\tau)} \, \mathrm{d}\tau \tag{272}$$

References